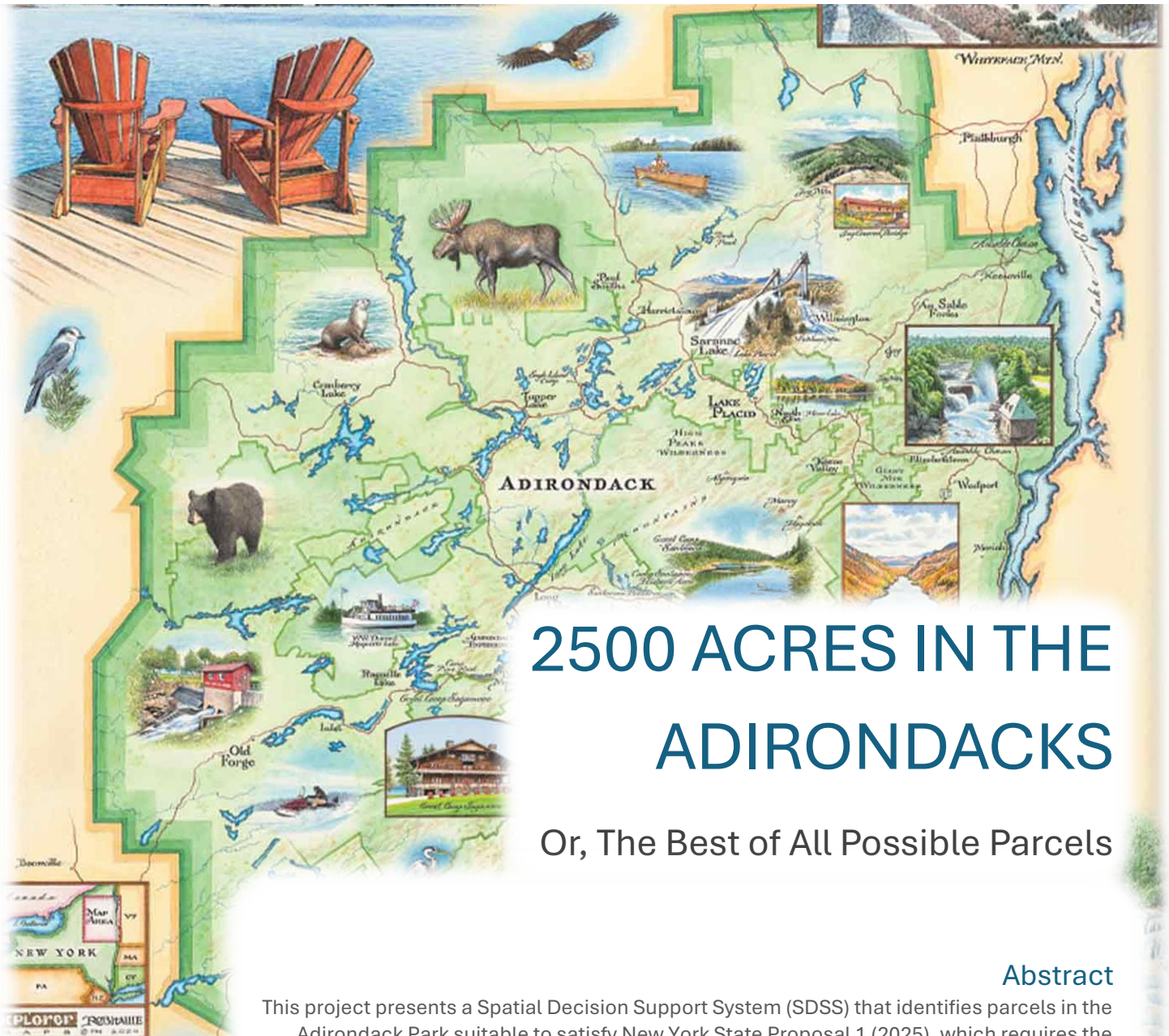




2500 ACRES IN THE ADIRONDACKS

Or, The Best of All Possible Parcels

Abstract

This project presents a Spatial Decision Support System (SDSS) that identifies parcels in the Adirondack Park suitable to satisfy New York State Proposal 1 (2025), which requires the state to acquire 2,500 acres of “equal or greater” ecological value to replace Forest Preserve land used by the Mount Van Hoevenberg Olympic Sports Complex. Built in ArcGIS Pro, the model scores candidate parcels on slope and land use, riparian proximity, rare habitat presence, and conservation gaps, letting users weight each criterion to reflect different stakeholder priorities and generate ranked recommendations.

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GTECH732 Final Paper

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Executive Summary

In November of 2025, New Yorkers voted in droves in a historic election. Amid that record turnout, one ballot measure stood out from the other city-level measures: proposal 1. The proposal was about 323 acres on the Adirondack Park Forest Preserve, where the New York Olympic Regional Development Authority (ORDA) built part of the Mount Van Hoevenberg Olympic Sports Complex on the Forest Preserve, against the New York Constitution's Article XIV, Section 1 which provides that this land "shall be forever kept as wild forest lands."¹ The proposal would add an amendment to the article in the constitution and would retroactively allow the construction of the Nordic skiing and biathlon trails and requires the state to purchase 2,500 new acres of land of "equal to or greater in value" in the Adirondack Park to add to the Forest Preserve, which the legislature must approve.

While the amendment did not define value, a Spatial Decision Support System (SDSS) could be able to measure the value of different parcels based on several criteria. A Senator would be able to, with recommendations from different experts, input weights, or importance, and receive recommendations based on their inputs. With those recommendations, they would be able to purchase the necessary parcels to comply with the amendment.

This SDSS is built upon a Multi-Criteria Evaluation (MCE) combined with geographic, topological, and administrative information that is publicly available, to be able to compute the value of a given set of parcels, based on ecological information and importance-weights (hereinafter referred to as inputs or weights) for each criterion. The final recommendations will be based on those inputs. The model evaluates parcels based on four distinct ecological criteria:

1. Land Use and Slope
2. Riparian Zones
3. Inclusion of Ecological Systems in which endangered species can survive (EDGE)
4. Inclusion of EDGE Systems which the DEC does not protect a lot of.

With those criteria, a user would be able to input a weight based on what factors they believe are most important and would receive a set of parcels which most fits their criteria best.

¹ New York State Senate, "Article XIV, Conservation."

The Mount Van Hoevenberg Olympic Sports Complex

Originally built in 1930 for the 1932 Winter Olympics in Lake Placid, NY, the sports complex has a 1455m long track built for bobsleigh, luge, and skeleton, and over 50km of trails for biathlons and cross-country skiing. The complex has since been used for training Olympic athletes and various competitions including the 1932 and 1980 Winter Olympics, the Winter Universiades (University Olympiads) in 1972 and 2023, and the Biathlon World Cup and Championships in 1987. The problem with the complex is that the biathlon and cross-country skiing trails were illegally built on 323 acres of land within the Adirondack Park Forest Preserve. The parcel itself is mostly covered by a mix of Deciduous and Evergreen Forest (NOAA 2021)² sits at an elevation of around 720 meters, with an average slope of 14 degrees. The parcel also sits at about 2000m east of the West Branch Ausable River. Now knowing this information, even if the legislation does not define value, we can look at the information on the parcel, and compare other available parcels to it, and measure whether they are of “equal or greater value”.

Literature Review

To quantify the ambiguous legal requirement of “equal or greater value,” this project relies on established environmental science and spatial modeling frameworks. The foundational methodology of this Spatial Decision Support System (SDSS) builds upon the Spatial Multi-Criteria Decision Analysis (S-MCDA) demonstrated by Kangas et al. (2016)³. In an assessment balancing winter sports infrastructure with biodiversity needs in Finland, the authors utilized a spatial grid system to calculate a Total Ecological Score (TES). This approach balances disparate environmental data, such as habitat rarity and state protection gaps, into a single, quantifiable surface, proving that spatial modeling can help resolve political and economic land-use disputes.

Beyond the mathematical framework, the specific ecological criteria evaluated by the model are grounded in established conservation literature. The overall scoring approach



Figure 1 - 12 Feb 1980: Barbara Petzold (R) of East Germany on her way to gold in the 10-km cross-country nordic skiing event during the Winter Olympics Games in Lake Placid, NY, USA. \ Mandatory Credit: Tony Duffy /Allsport

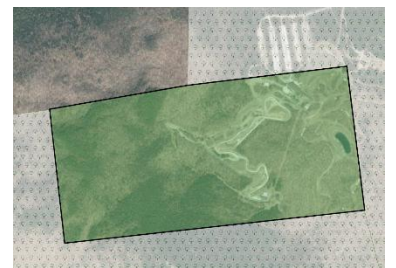


Figure 2 – Screenshot from ArcGIS showing the cross-country skiing and biathlon trails (green) cut out from the DEC land (little trees)

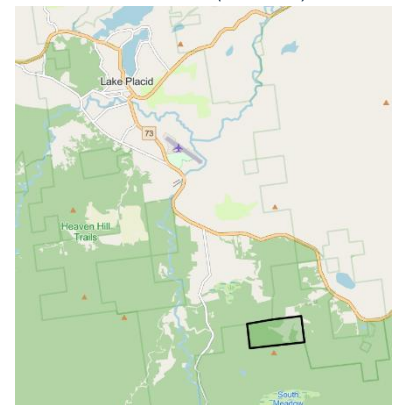


Figure 3 – Screenshot from ArcGIS showing location of parcel relative to Lake Placid

² NOAA Coastal Change Analysis Program, “Coastal Imagery Land Cover.”

³ Kangas et al., “A Method for Assessing Ecological Values to Reconcile Multiple Land Use Needs.”

(measuring each factor in its own units, normalizing it, and only then combining the factors by weight) follows Li and Nigh (2011)⁴, who used the same normalize-then-weight structure to prioritize private conservation parcels. To evaluate topographic viability, rather than a hard slope cutoff the model uses a continuous slope score, following Store and Kangas (2001)⁵, who argue that continuous membership functions preserve information that hard classification throws away; the score reaches zero at 50% slope, the upper limit documented for the spruce–northern hardwood forest community by the Maine Natural Areas Program. Watershed health is quantified with a slope-adjusted riparian buffer based on Wenger (1999)⁶ and corroborated by Mayer et al. (2005)⁷, which widens the protected buffer as the surrounding land steepens. Habitat rarity ranks are converted into scores using Rank-Order-Centroid weights (Barron and Barrett 1996)⁸, and the compactness of multi-parcel results follows the boundary-length logic of Marxan (Watts et al., 2009)⁹. By defining value through these established frameworks, the SDSS can evaluate a parcel's environmental merit. However, prioritizing which of these ecological factors is most important ultimately depends on the differing goals of the groups involved in the acquisition process. One important reason why we should want to input weights, as opposed to keeping all factors equally important, is that the SDSS does not account for time and climate change, and it does not take into account what is a high future risk.

The Stakeholders

This project identifies 4 major stakeholders that may influence our decision. First, various Conservation Groups suggest various metrics that they believe would make the replacement parcel one with a high ecological value. The second stakeholder would be the New York State Department of Environmental Conservation (NYS DEC or DEC) who do not want to purchase a parcel that is not connected to their current inventory of Wild Forests as a detached parcel would be harder to manage, and would lead to a fragmented forest which is less effective at environmental conservation than a connected forest. The New York State taxpayers would be the

⁴ Li and Nigh, “GIS-Based Prioritization of Private Land Parcels for Biodiversity Conservation: A Case Study from the Current and Eleven Point Conservation Opportunity Areas, Missouri.”

⁵ Store and Kangas, “Integrating Spatial Multi-Criteria Evaluation and Expert Knowledge for GIS-Based Habitat Suitability Modelling.”

⁶ Wenger, “A Review of the Scientific Literature on Riparian Buffer Width, Extent and Vegetation.”

⁷ Mayer et al., “Riparian Buffer Width, Vegetative Cover, and Nitrogen Removal Effectiveness: A Review of Current Science and Regulations.”

⁸ Barron and Barrett, “Decision Quality Using Ranked Attribute Weights.”

⁹ Watts et al., “Marxan with Zones: Software for Optimal Conservation Based Land- and Sea-Use Zoning.”

third stakeholder group, as we taxpayers want to make sure that our hard-earned tax dollars are put to good use, and getting the best value for each dollar spent (good cost-benefit ratio). Lastly, the Senate of the State of New York needs to comply with the passed amendment and have an easy-to-use tool to be able to find the best parcel to purchase.

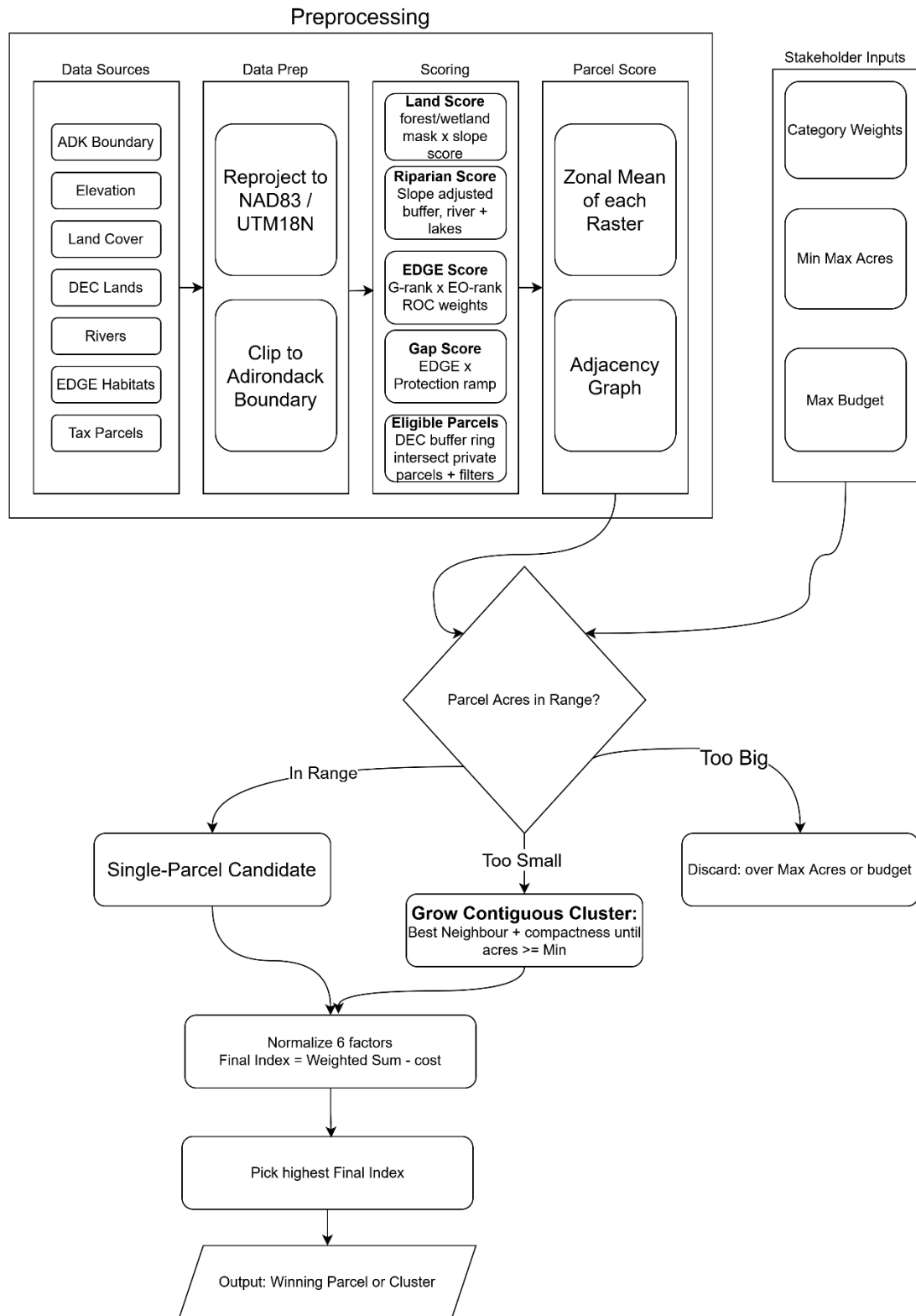
Stakeholder	Primary Objective	Metric
Conservation Groups	Ecological Integrity	Slope
		Land Cover Class
		EDGE Species Habitat
		Riparian Connectivity
NYS DEC	Management Efficiency	Forest Preserve Contiguity
Taxpayers	Fiscal Responsibility	Acquisition Cost
		No Standing Infrastructure
NYS Government	Compliance	Follows the amendment
	Ease of Use	Simple I/O

Conceptual Model

Blueprint

The goal of this model is to give parcels scores based on various hard coded and input criteria. The hard coded side of the model would be the criteria that have to be met before a parcel is even considered for selection, such as its adjacency to existing DEC lands, the type of land it sits on (such as developed vs undeveloped), and some basic requirements that would make a parcel ecologically viable. Once a parcel passes these tests, the model will use the user input weight to calculate the final score for the separate categories a parcel can be scored on, before ranking them and finding the parcel that best fits the user's importance-weights. This ensures that the final outputs are restricted to parcels with proven ecological value for the Adirondack Forest Preserve, while providing the flexibility to evaluate those parcels through multiple stakeholder perspectives.

Visual Representation



Criteria Selection

These criteria were selected for various reasons. First, of course, is that they were easily available and easy to work with as the metadata was thorough and came from reliable sources. Secondly, they represent a wide variety of ways to measure ecological value. While land can be measured in a myriad of ways, either by how much of a return on its investment you can make such as purchasing real estate, or by its potential for timber harvesting. But since the primary goal of this acquisition is to expand a protected preserve, we must measure value purely through an ecological lens. Even then, conservation valuation can be incredibly complex. By focusing the SDSS on four distinct categories, the model avoids relying on a single, isolated metric. Instead, it provides a broader ecosystem evaluation: balancing physical land viability (slope and land cover), watershed health (riparian zones), biodiversity (EDGE habitats), and prioritizing under-conserved ecosystems (gap analysis). This ensures the selected parcel is an ecologically important and fitting addition to the Adirondack Park's Forest Preserve.

Methodology & Implementation

Data Sources

Name (Nickname)	Source	Type	Description
Adirondack Boundary (ADKBound)	NYS Adirondack Park Agency	Polygon	Boundary of the Adirondack
Coastal Imagery Land Cover (LandUse)	NOAA Coastal Change Analysis Program	Raster	30m, represents Land Class values per cell
Index of Digital Elevation Models (DEM), New York (DEM)	Cornell University Geospatial Information Repository, U.S. Geological Survey	Raster	10m DEM
New York State Natural Heritage Community (EDGE)	New York State Department of Environmental Conservation	Polygon	Spatial extent of different level of significant, rare, high-quality natural communities and habitats

Adirondack Rivers System (Rivers)	NYS Adirondack Park Agency	Polygon	Represents river systems in the ADK
NYS Tax Parcels Public (Parcels)	NYS GIS Clearinghouse	Polygon	Represents all tax parcel information in counties which publish this data publicly (does not represent all counties which intersect with the ADK)
NYS DEC Land (DEC)	New York State Department of Environmental Conservation	Polygon	Represents parcels owned by the NYS DEC

Analysis Steps

Data Preparation

The first step in any GIS project, after gathering our data, is to normalize and clip it to our extent. A lot of the data sources apply to the entirety of the state of New York, which is outside of our search area of the Adirondacks. Since we are searching for a parcel within the Adirondacks, the data is first projected to NAD83 / UTM Zone 18N, which is a metric projected coordinate reference system that minimizes area and distance distortion across the northeastern United States, ensuring that acreage calculations and buffer distances are geometrically accurate rather than degree-based approximations. Once all of our data sources are reprojected and clipped/masked to the extent of the Adirondacks, the rest of the SDSS will be able to run all geoprocessing on smaller inputs, making the tool a lot faster.

Preliminary parcel filtering and Raster Scoring

Next, before we start selecting parcels based on their attributes, we need to filter out parcels based on their location relative to parcels owned by the DEC, and we need to calculate the value of a parcel based on its underlying topological and material properties. The two halves of the analysis, before the selection, are the topological rules (parcels adjacent to DEC lands), and topographic rules (scoring based on land use and cover). Then after the selection of parcels we want to consider, we pass the weights assigned by the user to the calculated score of each type of topographical variables and then assign each

parcel a final score based on the mean score of the value raster created by the topographical variables.

Final Parcel selection

Once each candidate receives a final score, we sort them by the highest and output the best result for the scenario: a single parcel, or a cluster of parcels grown together where no single one fits.

Procedure

The SDSS processes candidate parcels in four phases. First, each ecological raster: land use/slope, riparian proximity, EDGE habitat presence, and protection gaps, is reclassified and scored individually into its own 0–1 raster, without any stakeholder weight applied yet. Second, each raster is summarized onto every parcel via zonal statistics (mean coverage), so each parcel carries a per-acre intensity score for each factor. Third, for a given stakeholder those four ecological scores, along with parcel size and cost, are each normalized across the candidate set and combined into a single Final Index by a weighted sum; when no single parcel fits the requested acreage, adjacent parcels are grown into a contiguous cluster to meet it. Finally, the best parcel or cluster per scenario is kept, and a Python script iterates over every stakeholder scenario and combines their outputs into a single labeled feature class.

Logic

Parcel filtering: DEC-owned lands are dissolved into a single feature, reprojected to NAD83/UTM18N, and buffered 500m; the DEC footprint itself is then erased from that buffer to leave a thin buffer ring. Tax parcels are reprojected, restricted to those inside the ADK boundary under private ownership (property code 8), and selected for intersection with the DEC buffer ring — leaving only privately held parcels directly adjacent to existing DEC holdings.

Raster scoring: Land cover is reclassified into a binary mask of eligible forested and wetland classes, then multiplied by a continuous slope score. Rather than a hard cutoff, the slope score holds a full value across gentle-to-moderate terrain (0–15°) and decays linearly to zero at 26.57° (50% grade), the upper slope limit documented for the spruce–northern hardwood community by the Maine Natural Areas Program. Rivers and lakes are scored by a slope-adjusted riparian buffer following Wenger (1999)¹⁰: a base width of 100 ft plus 2 ft per 1% of slope, capped at 100 m, with the score decaying linearly from the water's

¹⁰ Wenger, “A Review of the Scientific Literature on Riparian Buffer Width, Extent and Vegetation.”

edge. EDGE natural-heritage communities are scored by combining a global rarity rank (G-rank) with an element-occurrence quality rank, each converted from ordinal ranks to cardinal values using Rank-Order-Centroid weights, then weighted dynamically by how much of that habitat type is already protected on existing DEC land: the boost ramps continuously from 2.0× where none is protected down to 1.0× at New York's legislated 30-by-30 protection target. Each factor is kept as its own scored raster rather than pre-combined, so that stakeholder weights can be applied later at the parcel level.

Final selection: Zonal statistics translate each score raster into a mean intensity score for every candidate parcel. For a given stakeholder, the four ecological scores plus parcel size (acres) and cost (market value) are each min-max normalized across the candidate set, then combined into a Final Index = $W_{\text{land}} \cdot \text{Land} + W_{\text{bio}} \cdot \text{EDGE} + W_{\text{rip}} \cdot \text{Riparian} + W_{\text{gap}} \cdot \text{Systems} + W_{\text{size}} \cdot \text{Size} - W_{\text{cost}} \cdot \text{Cost}$. Because the ecological scores are measured per acre, parcel size only enters through its own weight, keeping the six criteria independent. When no single eligible parcel falls within the requested acreage range, the model grows a contiguous cluster from a seed parcel, adding the best-scoring neighbor at each step and favoring parcels that fill in the cluster's shape over ones that stretch out a thin arm, which keeps the result compact. The single best-scoring parcel or cluster is exported as the scenario's output.

Stakeholder iteration: A CSV of named scenarios, one row per stakeholder weighting (land, biodiversity, riparian, gap, size, and cost weights, plus acreage and budget limits), drives the model in a loop, re-running the full scoring and ranking process for each scenario. A Python script then merges all scenario outputs into a single feature class, stamped with the scenario name, for side-by-side comparison.

Software Used

I used ArcGIS to do all my spatial analysis and model building, and I used Jupyter Notebook to run my python script to download the DEM data. I used the following python libraries within my code: os, time, requests, geopandas, csv, collections, math, and arcpy (including its Spatial Analyst module).

Results

Scenarios

Since the SDSS is an interactive one, the output might change depending on the input, so the following results are based on the following scenarios:

Run_Name	Land	Bio	Rip	Gap	Size	Cost	Max_Budget
Cheap	1	1	1	1	1	4	\$2,500,000
Eco	4	4	4	4	1	1	\$3,000,000

And for the global parameters I set them to:

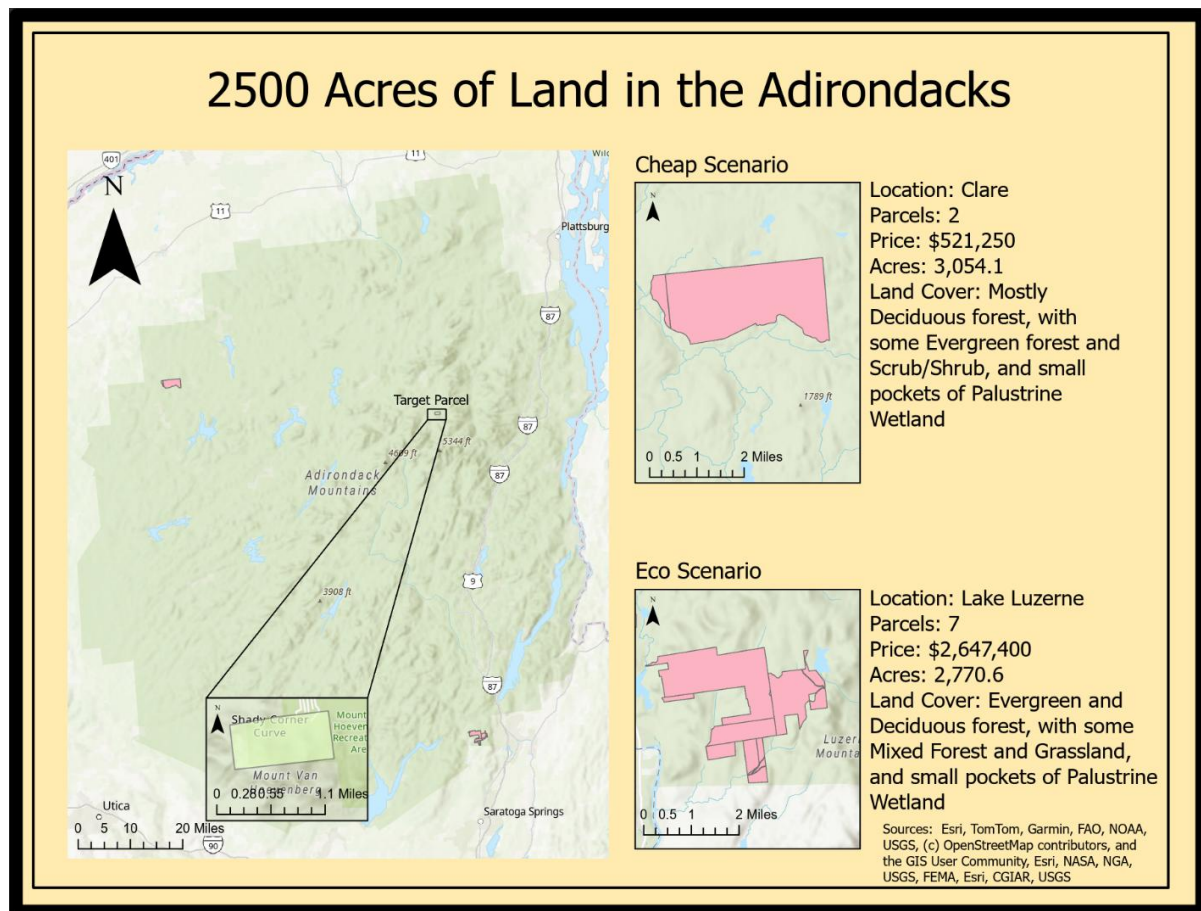
Min Acres: 2500

Max Acres: 4000

Num_Parcel: 1

This will make the SDSS run twice, once per scenario (Run_Name), where each parcel's factor scores are normalized and combined using that scenario's weights. The Size weight is held at 1 for both runs so the comparison isolates the cost-versus-ecology trade-off. For both runs it will only return the top result (Num_Parcel) between (inclusive) 2500 and 4000 acres: a single parcel where one fits, or a contiguous cluster of parcels grown to reach that range.

Output



Here we can see how the different weights gave us different parcels. The “Cheap” Scenario prioritized finding a cheap plot and gave us a 3,054-acre pair of adjacent parcels in the town of Clare, St. Lawrence County, worth \$521,250 and largely covered by deciduous forest with a few pockets of palustrine wetland (including Oclare Swamp). The Eco scenario, which weighted ecological criteria far more heavily than cost, returned a 2,771-acre contiguous cluster of seven parcels in Lake Luzerne, Warren County, covered by a mix of evergreen, deciduous, and mixed forest, priced at \$2,647,400. Notably, because the model now enforces the budget as a hard cap, the Eco result stays under its \$3,000,000 limit rather than exceeding it, and because it can assemble a contiguous cluster it is not limited to a single large parcel. This contrast illustrates how the SDSS surfaces fundamentally different trade-offs depending on which stakeholder priorities are weighted most heavily.

Discussion & Reflection

Critique

While building this SDSS, I identified a few limitations worth noting. The biggest hole is the lack of tax data for over half of the Adirondacks, a lot of which is the location of a good amount of our state's EDGE habitats. Although I found a feature layer which indicated all publicly available land, and could have subtracted that from the Adirondack polygon, I would not have been able to get the exact parcels and their value and acreage; at best, I could have gotten the general areas to search. I would also like to read more literature pertaining to industrial sites, and whether getting parcels near them to protect them is viable (and cost effective), or if we should focus on areas further away. I also did not do enough research on the specific act that protects our forests, and whether it includes a budget for restoration of newly acquired land and its maintenance (from hazardous material and invasive species removal). On the modeling side, the cluster-building step is greedy, so it finds a strong contiguous group of parcels but not necessarily the single best possible one, and the stakeholder weights are illustrative rather than elicited from real stakeholders through a formal method like the Analytic Hierarchy Process. Finally, while the model scores both rivers and lakes for riparian value, wetlands only enter through the land-cover criterion rather than being buffered as their own water feature, so some watershed connectivity may be underrepresented.

Q&A Synthesis

During the presentation, it was raised that the potential inclusion of the cost of demolishing standing structures, which I was hoping to add in the future to “free up” parcels that have higher ecological value than parcels with no standing structures but with lower ecological value. After more consideration, calculating the cost of an operation like this would be hard due to the lack of data, especially around long term affects the structure and systems could have had on a plot that would need to be taken care of before being able to convert the parcel to wild forest requirements.

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